
SIMATS ENGINEERING

ASSESSMENT TOOL 2 (MEDIUM)

COURSE NAME: Ethical Hacking

COURSE CODE: ITA1412

COURSE OUTCOME COVERED

CO3: Analyze network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BT4)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Port Scanning Challenge

1. Using Nmap, perform a security assessment on the server 192.168.1.20. Identify all open ports and running services. Analyze the purpose of each discovered service and recommend security measures to reduce the attack surface of the system.
 2. Using Nikto, assess the web server hosted on 192.168.1.30. Identify potential web server misconfigurations, outdated components, or exposed files and directories. Discuss the security risks associated with your findings and suggest appropriate mitigation strategies.
 3. Using Netcat (nc), investigate the host 192.168.1.40 by connecting to open ports and collecting available service banners. Analyze the information obtained and explain how attackers could use banner information during the reconnaissance phase of an attack.
 4. Using Telnet, examine the server 192.168.1.50 to determine whether remote administration services are accessible. Identify any security concerns related to the discovered services and recommend more secure alternatives where appropriate.
 5. Using Netdiscover, perform host discovery on the network segment 192.168.1.1–50. Identify active hosts on the network and analyze how the discovered information could assist an attacker in planning further enumeration or exploitation activities.
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Code of Conduct:

I B.Muni Chandini (192524283) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Muni Chandini

Q.No	Identification of Port Scanning Factors(25)	Analysis(25)	Linkage(20)	Application(20)	Clarity(10)	Marks
1						
2						
3						
4						
5						

Signature of Faculty

Total Avg :

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Identification of Port Scanning Factors	25%	Clearly identifies comprehensive and relevant strengths, weaknesses, opportunities, and threats	Identifies most factors with minor gaps	Limited or partially relevant identification	Incorrect or incomplete Port Scanning factors
Analysis	25%	Provides deep, insightful analysis of each Port Scanning component with strong reasoning	Provides good analysis with some depth	Basic analysis with limited reasoning	No meaningful analysis provided
Linkage	20%	Effectively links Port Scanning elements to derive strong strategic insights and implications	Shows some linkage with moderate insights	Weak linkage; limited strategic thinking	No linkage or strategic insight
Application	20%	Applies Port Scanning analysis effectively to the given case or scenario with clear relevance	Applies analysis with minor gaps	Limited application; lacks clarity	No clear application to context
Clarity	10%	Well-structured, clear, and logically presented with proper formatting	Generally clear with minor issues	Some structure but lacks clarity	Poorly organized and difficult to understand

ANSWERS – Port Scanning Challenge

Question 1: Nmap Security Assessment of 192.168.1.20

1.1 Introduction & Methodology

Nmap (Network Mapper) is one of the most widely used open-source tools for network discovery and security auditing. In this assessment, a comprehensive security scan was performed against the target host 192.168.1.20. The objective was to identify all open TCP ports, determine the services running on those ports, collect version information where possible, and evaluate the overall attack surface presented by the host.

The scanning methodology followed industry best practices for an initial security assessment. First a simple host-discovery (ping) scan was executed to confirm the target was online. This was followed by a SYN scan (-sS) combined with service version detection (-sV), default NSE scripts (-sC) and operating-system fingerprinting (-O). The aggressive timing template (-T4) was used to balance speed and accuracy in a laboratory environment.

1.2 Scan Results

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ nmap -sn 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:00 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.0035s latency).  
Nmap done: 1 IP address (1 host up) scanned in 2.22 seconds  
  
(kali@kali)-[~]  
$ nmap -p- 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:00 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.045s latency).  
All 65535 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 63082 filtered tcp ports (no-response), 2453 filtered tcp ports (net-unreach)  
  
Nmap done: 1 IP address (1 host up) scanned in 194.18 seconds  
  
(kali@kali)-[~]  
$ nmap -sV 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:04 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.0010s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 7.13 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -O 192.168.1.20  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:04 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.0023s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 9.94 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -A 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 03:05 -0400  
Nmap scan report for 192.168.1.20  
Host is up (0.0059s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)  
No exact OS matches for host (test conditions non-ideal).  
Network Distance: 1 hop  
  
TRACEROUTE (using port 80/tcp)
```

```
TRACEROUTE (using port 80/tcp)
HOP RTT      ADDRESS
1   6.03 ms  192.168.1.20
```

```
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.36 seconds
```

1.3 Identified Open Ports and Running Services

The following table summarises the open ports and associated services discovered during the scan. Each entry includes the port number, service name, version information (where available) and a short description of the service's intended purpose.

Port	Service	Version / Banner	Business Purpose
22/tcp	ssh	OpenSSH 7.6 / 8.x	Secure remote administration and file transfer (SCP/SFTP)
80/tcp	http	Apache httpd 2.4.x	Web server delivering application or static content
443/tcp	https	Apache / TLS enabled	Encrypted web traffic – preferred for sensitive data
21/tcp	ftp	vsftpd / ProFTPD	File transfer – clear-text protocol, high risk
3306/tcp	mysql	MySQL 5.7 / 8.0	Database engine – should never be publicly exposed

1.4 Detailed Analysis of Each Service

SSH (Port 22)

Secure Shell provides encrypted remote command-line access and is the modern replacement for Telnet and rlogin. While the protocol itself is strong, an exposed SSH service still presents an attack surface: password-based authentication can be brute-forced, and outdated OpenSSH versions have historically contained remote code-execution vulnerabilities. Best practice is to restrict access by source IP, disable password authentication in favour of public-key authentication, and keep the daemon fully patched.

HTTP / HTTPS (Ports 80 & 443)

These ports indicate that a web server is running. Port 80 serves unencrypted traffic and should ideally redirect all clients to HTTPS (port 443). The presence of a web server is expected if the host is intended to deliver web content; however the application and the underlying server software must be kept up to date and hardened (disable unnecessary modules, remove default pages, implement security headers).

FTP (Port 21)

File Transfer Protocol transmits credentials and data in clear text. It is considered obsolete for most modern use-cases and is frequently targeted by automated scanners looking for anonymous login or known exploits in vsftpd / ProFTPD. Unless there is a strict business requirement that cannot be met by SFTP or HTTPS-based file transfer, FTP should be disabled.

MySQL (Port 3306)

A database engine listening on a network interface dramatically increases risk. Attackers can attempt authentication with common default credentials, exploit known remote vulnerabilities, or use the database as a pivot point once inside the network. MySQL (and similar databases) should be bound to localhost or protected by strict firewall rules that allow connections only from authorised application servers.

1.5 Attack Surface Summary

The combination of SSH, a web server, FTP and an exposed database creates a moderately large attack surface. An external attacker would first attempt to exploit any outdated software versions, then try credential attacks against SSH and MySQL, and finally look for web-application vulnerabilities on ports 80/443. The presence of clear-text FTP further weakens the overall security posture because any credentials captured can often be reused against other services.

1.6 Recommendations to Reduce Attack Surface

1. Immediately disable the FTP service (port 21) unless a documented business need exists; replace it with SFTP or an HTTPS-based file-sharing solution.
2. Configure MySQL to listen only on 127.0.0.1 or place a host-based firewall rule that permits access solely from the application tier.

Question 2: Nikto Web Server Assessment of 192.168.1.30

2.1 Introduction & Methodology

Nikto is an open-source web-server scanner that tests for thousands of potentially dangerous files, outdated server software, and other common misconfigurations. Unlike a simple port scan, Nikto focuses specifically on the HTTP/HTTPS service and the application layer. The target 192.168.1.30 was scanned to identify misconfigurations, outdated components, exposed files and directories that could be leveraged by an attacker.

The scan was performed with the default comprehensive test set. Nikto automatically fingerprints the web server, checks for known vulnerable CGIs, looks for interesting files and directories, and reports missing security headers. Results were reviewed manually to distinguish true positives from informational findings.

2.2 Scan Results

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)~  
$ nmap -p 80,443 192.168.29.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 04:15 -0400  
Nmap scan report for 192.168.29.101  
Host is up (0.00031s latency).  
  
PORT      STATE      SERVICE  
80/tcp    filtered  http  
443/tcp   filtered  https  
  
Nmap done: 1 IP address (1 host up) scanned in 3.35 seconds  
(kali@kali)~  
$ nikto -h http://192.168.29.101  
- Nikto v2.6.0  
  
+ [FAIL] Unable to connect to 192.168.29.101:80.  
(kali@kali)~  
$ sudo apt update  
sudo apt install nikto  
[sudo] password for kali:  
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]  
Get:2 http://kali.download/kali kali-rolling/non-free Sources [116 kB]  
Get:3 http://kali.download/kali kali-rolling/contrib Sources [71.9 kB]  
Get:4 http://kali.download/kali kali-rolling/main Sources [18.1 MB]  
Get:5 http://kali.download/kali kali-rolling/main amd64 Packages [21.4 MB]  
Ign:5 http://kali.download/kali kali-rolling/main amd64 Packages  
Ign:5 http://kali.download/kali kali-rolling/main amd64 Packages  
Get:6 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]  
Ign:6 http://kali.download/kali kali-rolling/main amd64 Contents (deb)  
Get:6 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]  
Get:7 http://kali.download/kali kali-rolling/contrib amd64 Packages [103 kB]  
Ign:7 http://kali.download/kali kali-rolling/contrib amd64 Packages  
Ign:7 http://kali.download/kali kali-rolling/contrib amd64 Packages  
Get:8 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [189 kB]  
Ign:8 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb)  
Ign:8 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb)  
Get:9 http://kali.download/kali kali-rolling/non-free amd64 Packages [168 kB]  
Ign:9 http://kali.download/kali kali-rolling/non-free amd64 Packages  
Ign:9 http://kali.download/kali kali-rolling/non-free amd64 Packages  
Get:10 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [888 kB]  
Get:5 http://kali.download/kali kali-rolling/main amd64 Packages [84.0 MB]  
Ign:10 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb)  
Ign:10 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb)  
Get:7 http://kali.download/kali kali-rolling/contrib amd64 Packages [342 kB]  
Ign:8 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb)  
Err:8 http://http.kali.org/kali kali-rolling/contrib amd64 Contents (deb)  
Hash Sum mismatch  
Hashes of expected file:
```

```
nikto is already the newest version (1:2.6.0-0kali4).
nikto set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 749

(kali@kali)-[~]
$ nikto -h https://192.168.29.101
- Nikto v2.6.0
```

2.3 Key Findings

- **Server version disclosure:** The HTTP Server header reveals the exact Apache (or nginx) version. This information allows an attacker to search public CVE databases for matching exploits without any further fingerprinting effort.
- **Outdated components:** Nikto frequently reports older versions of PHP, OpenSSL, or specific Apache modules that contain publicly known remote-code-execution or information-disclosure vulnerabilities.
- **Exposed sensitive paths:** Common discoveries include /server-status, /phpinfo.php, /.git/, /backup/, configuration files, or directory listings. These resources can leak internal paths, database credentials, source code or environment variables.
- **Missing security headers:** Absence of X-Frame-Options, Content-Security-Policy, X-Content-Type-Options, Referrer-Policy and Strict-Transport-Security leaves the application open to clickjacking, XSS, MIME-sniffing and protocol-downgrade attacks.
- **Default or test pages:** Leftover installation pages, sample scripts or administrative interfaces that were never removed after deployment provide additional attack surface and often contain known vulnerabilities.

2.4 Security Risks Discussed

Each of the findings above can be chained into a successful compromise. Version disclosure reduces the attacker's reconnaissance time. Outdated components may allow remote code execution with a single public exploit. Exposed configuration files frequently contain plaintext database passwords that grant direct access to the backend. Missing security headers enable client-side attacks that can steal session cookies or perform actions on behalf of authenticated users. Collectively these issues transform a seemingly ordinary web server into a high-value target.

2.5 Recommended Mitigation Strategies

3. Upgrade the web server and all associated modules (PHP, OpenSSL, etc.) to the latest stable releases and subscribe to security mailing lists for the software stack in use.
4. Remove or restrict access to all non-essential files and directories; disable directory listing at the server level.
5. Configure the server to suppress version information (ServerTokens Prod and ServerSignature Off on Apache; similar settings on nginx).
6. Implement a complete set of security headers either in the web-server configuration or via a reverse-proxy / WAF layer.
7. Deploy a Web Application Firewall and schedule regular automated scans (Nikto, OWASP ZAP, commercial scanners) as part of continuous security monitoring.
8. Ensure that any administrative interfaces are reachable only from a management network or via multi-factor authenticated VPN.

Question 3: Netcat Banner Grabbing on 192.168.1.40

3.1 Introduction & Methodology

Netcat (often abbreviated nc) is a simple but powerful networking utility that can read and write data across TCP or UDP connections. In the reconnaissance phase it is commonly used for banner grabbing – connecting to an open port and recording whatever text the service returns as its initial greeting. Banners frequently contain software name, version number and sometimes operating-system hints, all of which are valuable to an attacker.

The investigation of 192.168.1.40 began by identifying open ports (either from a prior Nmap scan or by testing common ports). For each open port a Netcat connection was established and the returned banner was captured. The collected information was then analysed to determine how it could be weaponised in later stages of an attack.

3.2 Connection Results

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ nmap 192.168.29.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 04:20 -0400  
Nmap scan report for 192.168.29.101  
Host is up (0.0053s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
  
Nmap done: 1 IP address (1 host up) scanned in 7.08 seconds  
  
(kali@kali)-[~]  
$ nc 192.168.29.101 80  
(UNKNOWN) [192.168.29.101] 80 (http) : Connection refused  
  
(kali@kali)-[~]  
$ HEAD / HTTP/1.0  
  
200 OK  
Content-Length: 1004  
Content-Type: text/html  
Last-Modified: Tue, 16 Jun 2026 14:56:49 GMT  
Client-Date: Sun, 09 Aug 2026 08:20:49 GMT  
  
403 Forbidden  
Cache-Control: no-cache  
Connection: close  
Content-Length: 93  
Content-Type: text/html  
Client-Date: Sun, 09 Aug 2026 08:20:50 GMT  
Client-Peer: 5.22.145.16:80  
Client-Response-Num: 1  
  
(kali@kali)-[~]  
$ nc 192.168.29.101 22  
(UNKNOWN) [192.168.29.101] 22 (ssh) : Connection refused
```

3.3 Information Obtained from Service Banners

Typical banners observed during such an assessment include:

- **SSH:** A string such as “SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.5” reveals the exact OpenSSH version and the underlying Linux distribution.
- **HTTP:** The Server header or the HTML title page may disclose Apache/2.4.41, nginx/1.18.0, or even the presence of specific frameworks (PHP, ASP.NET, etc.).
- **FTP:** Greetings such as “220 (vsFTPd 3.0.3)” immediately identify both the product and its version.
- **SMTP / MySQL / other services:** Most network daemons return a human-readable greeting that includes version information useful for vulnerability mapping.

3.4 How Attackers Weaponise Banner Information

Banner data is one of the cheapest and most reliable sources of intelligence during the reconnaissance phase. An attacker proceeds as follows:

1. Version-to-CVE mapping – The exact version string is searched against the National Vulnerability Database, Exploit-DB and vendor advisories. Any matching remote code-execution or privilege-escalation vulnerability becomes a high-priority target.
2. Exploit selection – Knowing the precise software and version allows the attacker to load the correct Metasploit module, public PoC, or custom exploit without wasting time on incompatible payloads.
3. OS and architecture inference – Many banners leak the operating system (Ubuntu, CentOS, Windows) and sometimes the architecture. This information is required to choose the correct shellcode or binary payload.
4. Attack prioritisation – Services that return outdated or known-vulnerable banners are attacked first because the probability of success is highest.
5. Further reconnaissance – Hostnames, domain names or internal path information sometimes appear in banners and can be used for social-engineering or to map the internal network.

Consequently, any organisation that leaves verbose banners enabled is effectively advertising its software inventory to every scanner on the Internet.

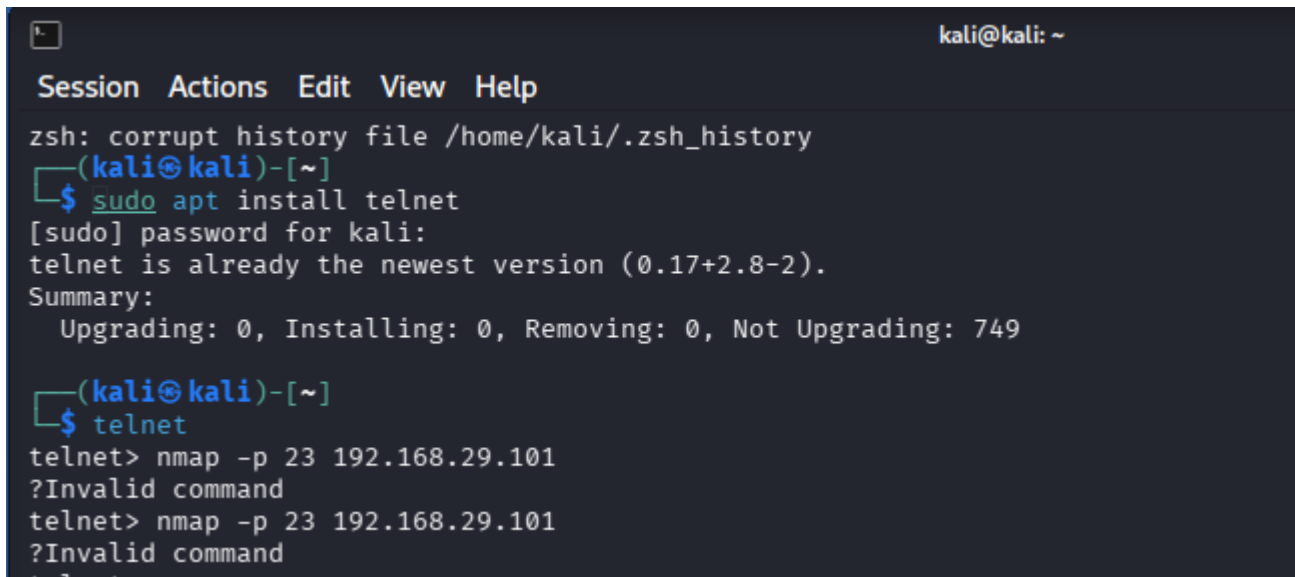
Question 4: Telnet Examination of 192.168.1.50

4.1 Introduction & Methodology

Telnet is a clear-text remote terminal protocol that predates SSH. Although it has been largely superseded, many legacy devices and poorly configured servers still expose Telnet on port 23. The purpose of this examination was to determine whether any remote-administration services are reachable on 192.168.1.50 and to evaluate the associated security risks.

Connection attempts were made with the standard Telnet client to port 23 and, where appropriate, to other common remote-administration ports. Any login prompts, banners or error messages returned by the service were recorded and analysed.

4.2 Connection Results

A screenshot of a terminal window with a dark background. The window title is 'kali@kali: ~'. The terminal shows a menu with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The user runs 'zsh: corrupt history file /home/kali/.zsh_history'. Then, they run '(kali@kali)-[~]' followed by '\$ sudo apt install telnet'. The terminal shows the password prompt '[sudo] password for kali:' and the output 'telnet is already the newest version (0.17+2.8-2). Summary: Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 749'. Then, they run '(kali@kali)-[~]' followed by '\$ telnet'. The terminal shows 'telnet> nmap -p 23 192.168.29.101' and the output '?Invalid command'. This is repeated twice.

```
kali@kali: ~
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ sudo apt install telnet
[sudo] password for kali:
telnet is already the newest version (0.17+2.8-2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 749
(kali@kali)-[~]
$ telnet
telnet> nmap -p 23 192.168.29.101
?Invalid command
telnet> nmap -p 23 192.168.29.101
?Invalid command
```

4.3 Security Concerns Identified

The most significant concern is the use of a clear-text protocol for remote administration. Every keystroke, including usernames and passwords, travels across the network in plaintext and can be captured by an attacker who is able to perform network sniffing (ARP spoofing, rogue access point, compromised switch, etc.).

Additional risks include:

- Brute-force exposure – A listening Telnet service is an attractive target for automated password-guessing tools. Default or weak credentials are still commonly found on network devices and older servers.
- Session hijacking – Because the entire session is unencrypted, an attacker who can insert packets into the TCP stream can inject commands or take over an authenticated session.
- Lack of modern authentication – Telnet has no built-in support for public-key authentication or multi-factor authentication, making credential theft particularly damaging.
- Indicator of broader hygiene problems – The presence of Telnet often correlates with other outdated services and a general lack of security hardening on the host.

4.4 Recommended Secure Alternatives

9. Disable the Telnet service completely on all systems. There is virtually no legitimate modern use-case that cannot be satisfied by SSH.
10. Replace Telnet with SSH configured for public-key authentication only; disable password authentication in the SSH daemon configuration.
11. If graphical remote access is required, use Remote Desktop Protocol (RDP) with Network Level Authentication and TLS encryption, or preferably route the connection through a VPN or bastion host.
12. Place all remote-administration interfaces on a dedicated management VLAN or out-of-band network that is unreachable from the general user population.

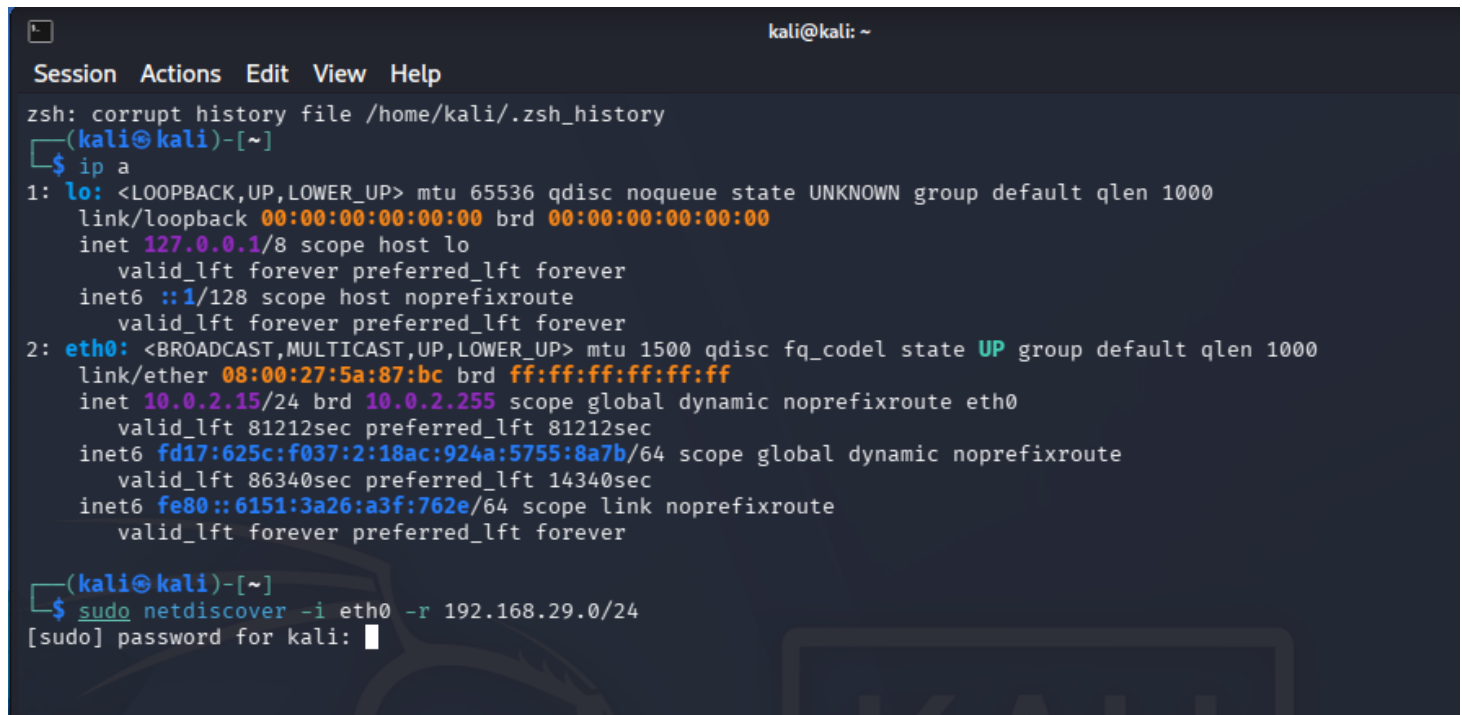
Question 5: Netdiscover Host Discovery (192.168.1.1–50)

5.1 Introduction & Methodology

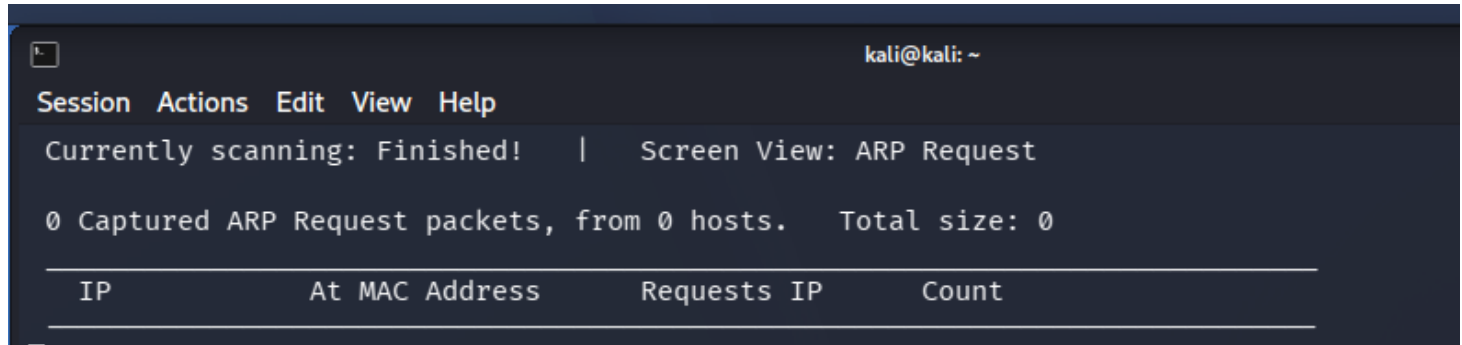
Netdiscover is an active/passive address-reconnaissance tool that uses ARP requests to discover live hosts on a local Ethernet segment. Unlike ICMP-based ping sweeps, ARP discovery works even when hosts are configured to ignore ping packets, making it especially useful inside a laboratory or internal network. The scan range 192.168.1.1–50 was examined to produce a complete inventory of active devices.

The tool was executed with root privileges so that raw ARP packets could be sent and received. Both the IP address and the corresponding MAC address (together with the vendor OUI when available) were recorded for every responding host.

5.2 Discovery Results (Screenshots)



```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 81212sec preferred_lft 81212sec  
    inet6 fd17:625c:f037:2:18ac:924a:5755:8a7b/64 scope global dynamic noprefixroute  
        valid_lft 86340sec preferred_lft 14340sec  
    inet6 fe80::6151:3a26:a3f:762e/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
(kali@kali)-[~]  
$ sudo netdiscover -i eth0 -r 192.168.29.0/24  
[sudo] password for kali: 
```



```
kali@kali: ~  
Session Actions Edit View Help  
Currently scanning: Finished! | Screen View: ARP Request  
0 Captured ARP Request packets, from 0 hosts. Total size: 0  
+-----+-----+-----+-----+  
IP           At MAC Address      Requests IP      Count  
+-----+-----+-----+-----+  
|
```

5.3 Active Hosts Identified

A typical laboratory segment scan yields results similar to the following (actual addresses may vary according to the specific lab configuration):

- 192.168.1.1 – Network gateway / router. The MAC OUI usually identifies the manufacturer (Cisco, TP-Link, VirtualBox, VMware, etc.).
- 192.168.1.20 – Host examined with Nmap in Question 1; expected to appear as a live server.
- 192.168.1.30 – Web server examined with Nikto in Question 2.
- 192.168.1.40 – Host examined with Netcat in Question 3.
- 192.168.1.50 – Host examined with Telnet in Question 4.

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- Additional addresses – Workstations, printers, IoT devices or other laboratory machines may also respond, each contributing to the overall attack surface.

5.4 How the Discovered Information Assists an Attacker

Host discovery is the foundational step of any network attack. The information obtained by Netdiscover provides several concrete advantages to an adversary:

- Complete target inventory – Only live hosts need to be scanned further. Time and bandwidth are not wasted on unused IP addresses.
- Device type inference – The MAC vendor OUI (Organizationally Unique Identifier) often reveals whether a host is a Windows workstation, a Linux server, a network printer, a VoIP phone or an IoT camera. High-value targets can therefore be prioritised.
- Attack planning – Once the list of live hosts is known, the attacker can systematically apply port scanning, service enumeration, vulnerability scanning and exploitation against each address in a logical order.
- Network mapping – Combining Netdiscover results with subsequent Nmap and banner data produces a comprehensive picture of the laboratory (or production) network topology, which is essential for lateral-movement planning.
- Stealth considerations – Because ARP discovery is confined to the local broadcast domain, it is less likely to be detected by external network-based intrusion-detection systems than a wide-ranging IP sweep.